

CSA1717 - Artificial Intelligence

Assessment Tool-3

Descriptive written test

Name: A Rohan

Reg No: 192572100

Course: Artificial Intelligence

Code: CSA1717

1. Intelligence Agents in a smart Home Automation system

1. Simple Reflex Agent

A simple reflex agent makes decision based only on the current sensor input using predefined If-THEN rules. It does not store past information or learn from experience.

Function in the smart Home

- if motion is detected - turn on the lights
- if room temperature exceeds 28°C → Turn on the air conditioner
- if a door opens while the alarm is armed → Activate the security alarm

Advantages

- fast response to real time agent
- easy to design and implement
- low computational cost

Limitations

- cannot remember previous event
- cannot adapt to changing user preference
- poor performance in complex situation

2. Model Based Agent

A Model-Based Agent maintains an internal model (memory) of the environment and updates it using current and past sensor information.

Function in the smart Home

- Remember whether people are inside the house
- Keep light on even if no movement is detected for short time
- tracks previous temperature reading to avoid frequent A/C switching
- Monitor whether doors and windows are locked

Advantages

- handles partially observable environments.
- makes more accurate decisions using memory
- improves system reliability

Limitation

- more complex than a simple reflex agent
- require additional memory and processing

3. Goal-Based Agent

A Goal-Based Agent selects actions that help achieve specific objectives

Goal in the smart Agent

- maintain room temperature at 24°C
- reduce electricity consumption
- Keep the home secure
- Ensure all light are turned off when the house is empty

Function

The agent evaluates different actions and chooses the one best achieves the desired goal

Ex: if the user leaves Home $\rightarrow$ Turn off lights

Advantages

- make intelligent long term decision
- suitable for planning multiple action

Limitation

- Requires more computation
- Does not directly compare different level of satisfaction

4. Utility Based Agent

- select the action with highest overall benefit
- Example: Balance comfort, energy efficiency and security
- Pros: Gives the best overall performance
- Cons: most complex

Best Approach

A hybrid agent (simple reflex + model based + goal-based + utility) is the most effective because it responds quickly to real time event

Conclusion: A hybrid intelligent agent provide the best balance of comfort efficiency and security

Robot Maze Problem using uniformed search

A robot is placed in an unknown maze and must find the exit without knowing the layout. It can use uniformed search algorithms such as uniform cost search (UCS) and iterative Deepening search (IDS) to explore the maze.

uniform cost search (UCS)

UCS expands the path with the lowest cumulative cost first. If different paths have different movement cost, UCS guarantees the optimal (least cost) path to the exit.

Advantages

- finds the optimal solution
- works well when path cost vary

Disadvantages

- high time complexity: $O(b^{l+1} \cdot l \cdot \epsilon \cdot 1)$
- high space complexity
- uses a large amount of memory

Iterative Deepening Search (IDS)

IDS repeatedly performs Depth Limited search with increasing depth limits until the goal is found. It combines the benefits of Depth first search and Breadth first search.

Advantages

- complete and optimal for equal step costs
- low memory usage
- space complexity $O(b \cdot d)$

Disadvantages

- Repeats searches at each depth
- time complexity: $O(b^d)$

Intelligent Agents

simple Reflex Agent: Reacts only to current surrounding
unintelligible for an unknown maze

Model Based Agent: maintain memory of explored paths
and visited location preventing repeated exploration

Goal Based Agent: searches for action that achieves
the goal of reaching the exit

Utility Based Agent: chooses paths that maximize efficiency
by considering distance, cost, and safety

Best Combination

The Model-Based Goal Based Agent with uniform cost
search (UCS) is most efficient choice. The Model Based
Agent remembers explored path when movement cost vary.
if all movements have equal cost and memory is limited
iterative Deepening search (IDS) is better choice because
it requires much less memory

Conclusion

for an unknown maze, UCS is best for finding
the optimal path, while IDS is preferred when memory
is limited. A Model-Based goal Based Agent combined
with UCS provides the most reliable and intelligence
solution for efficient maze navigation